

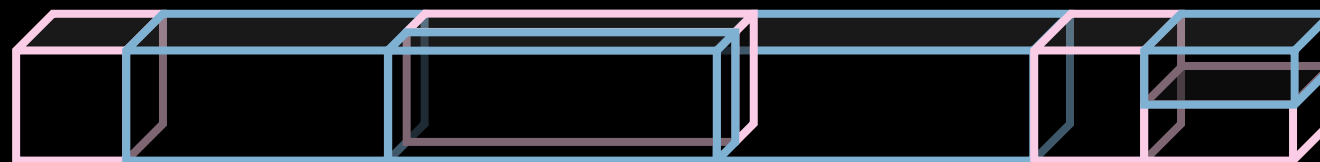
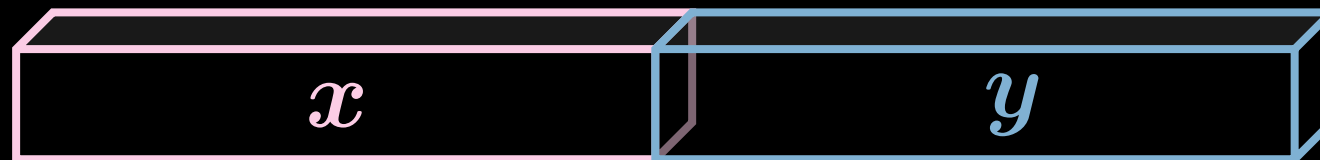
Energy-Based Self-Supervised Deep Learning

Capture dependencies, predicting everything from everything,
while handling multimodality and stochasticity

Self-supervised learning

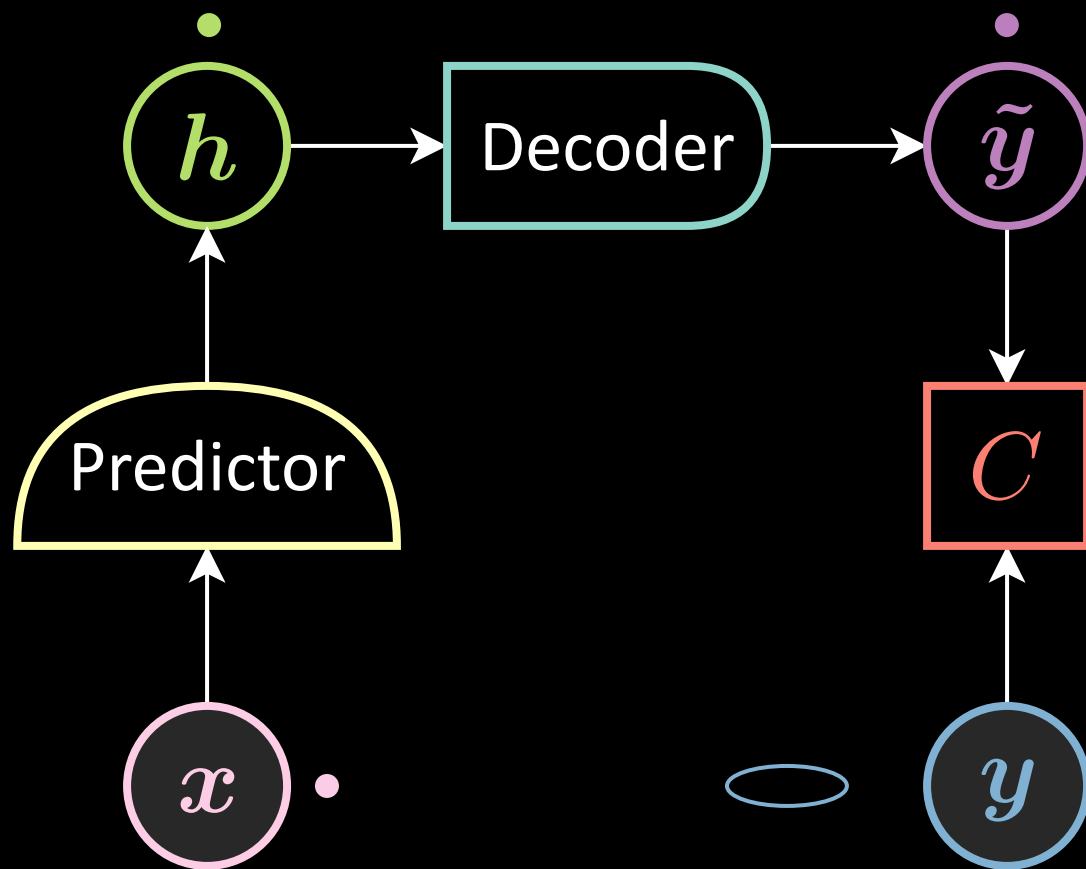
Predicting everything from everything

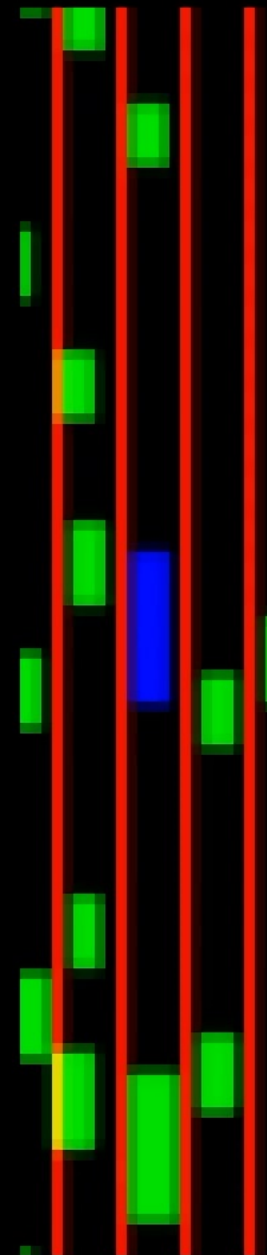
time or space



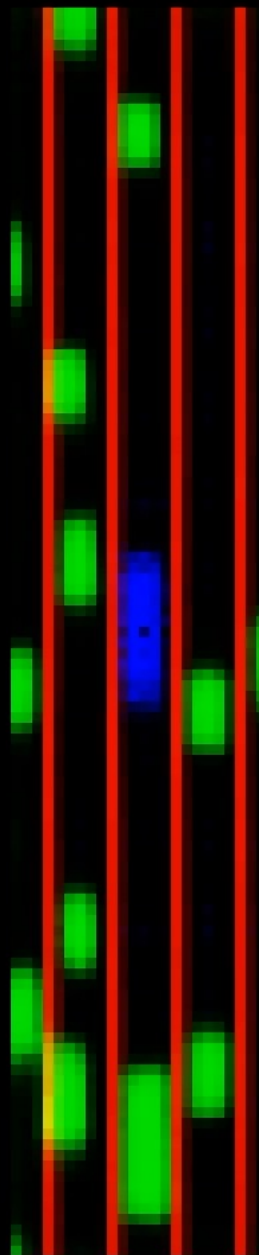
1. learning hierarchical representations of the world

2. learning predictive (forward) models of the world





Actual future



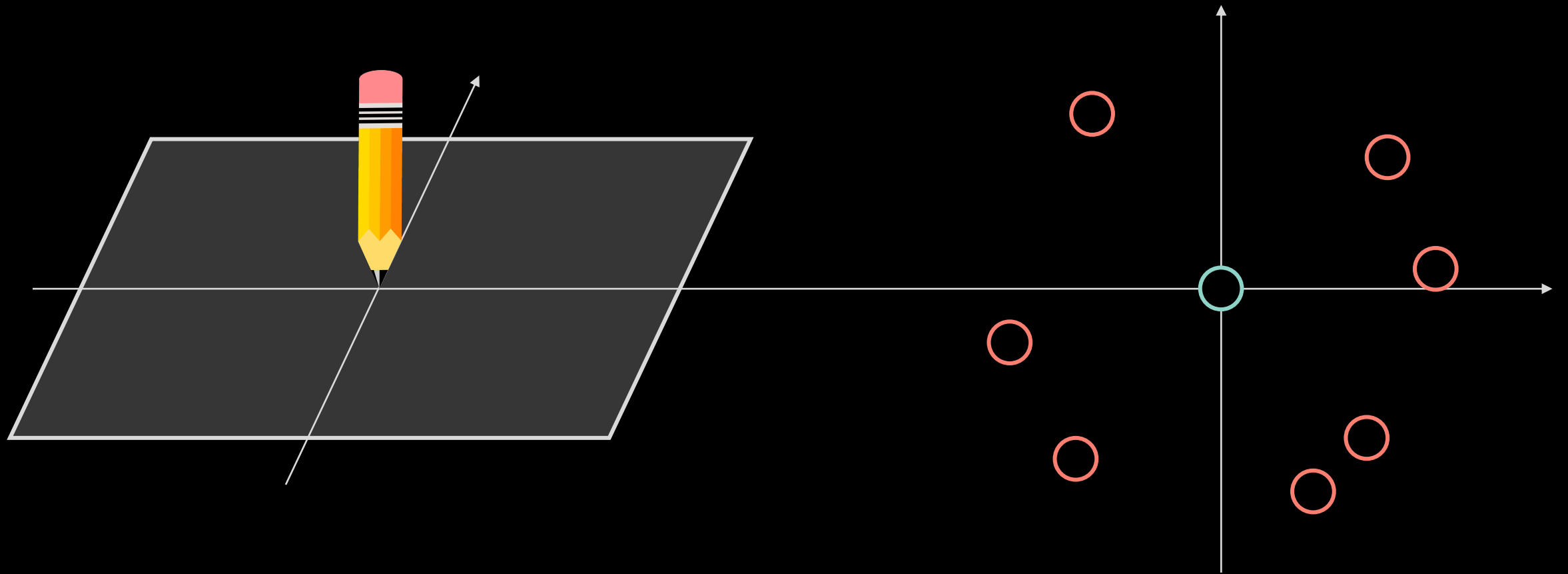
Deterministic

$t = 1$

direction of motion



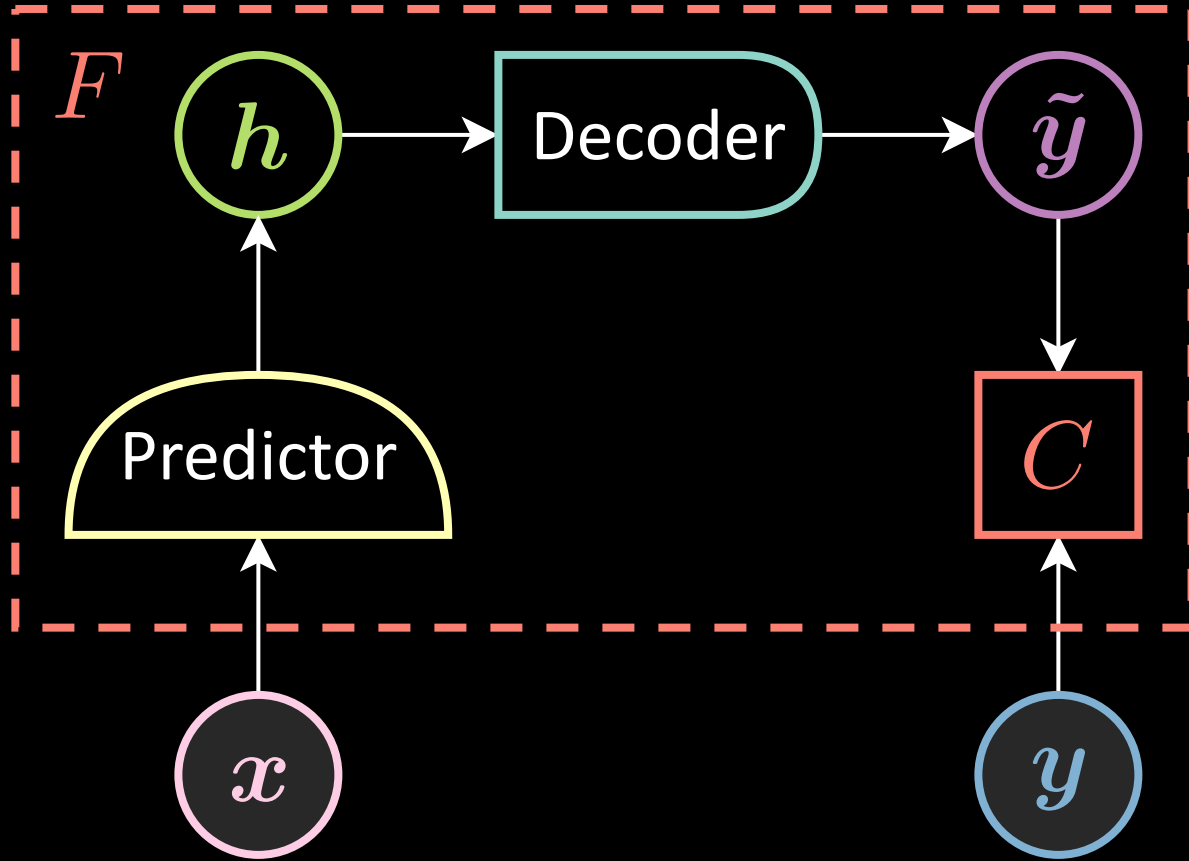
Future's multimodality / uncertainty



Energy-based models

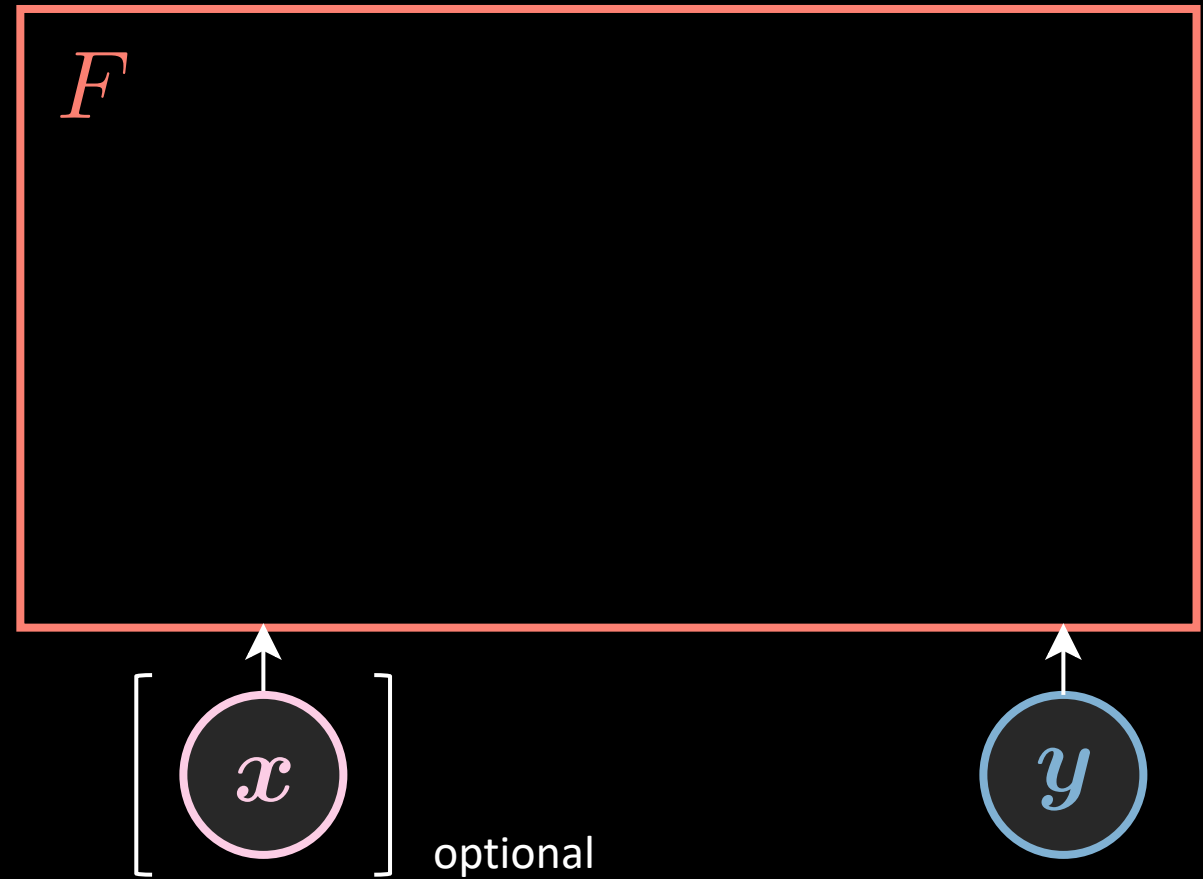
Capture dependencies through an energy function

Feed-forward: output produced in a finite nb of steps



$$F(x, y) = C(y, \tilde{y})$$

Energy model: inference through energy minimisation

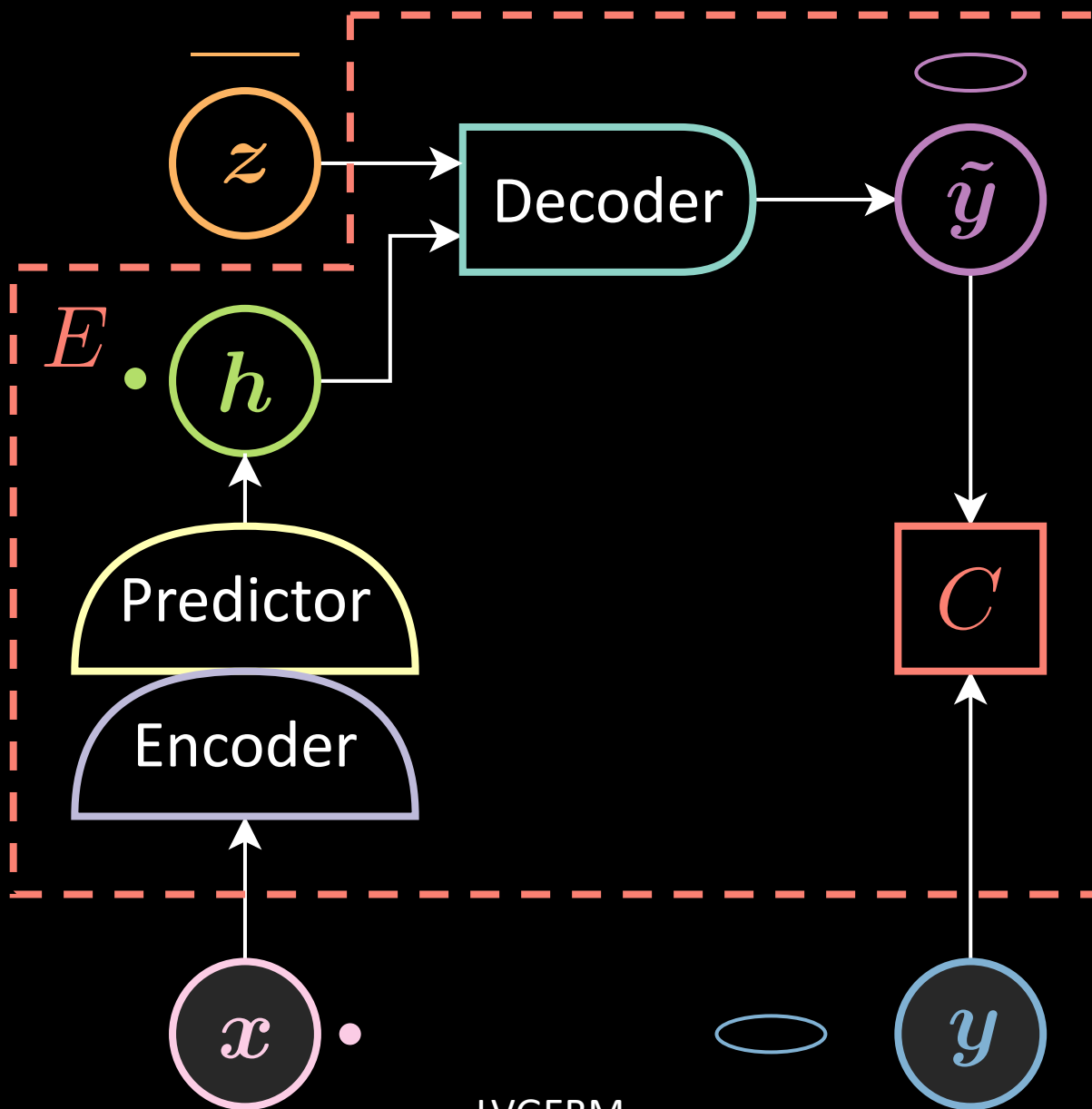


$$\check{y} = \arg \min_{y \in \mathcal{Y}} F(x, \mathcal{Y})$$

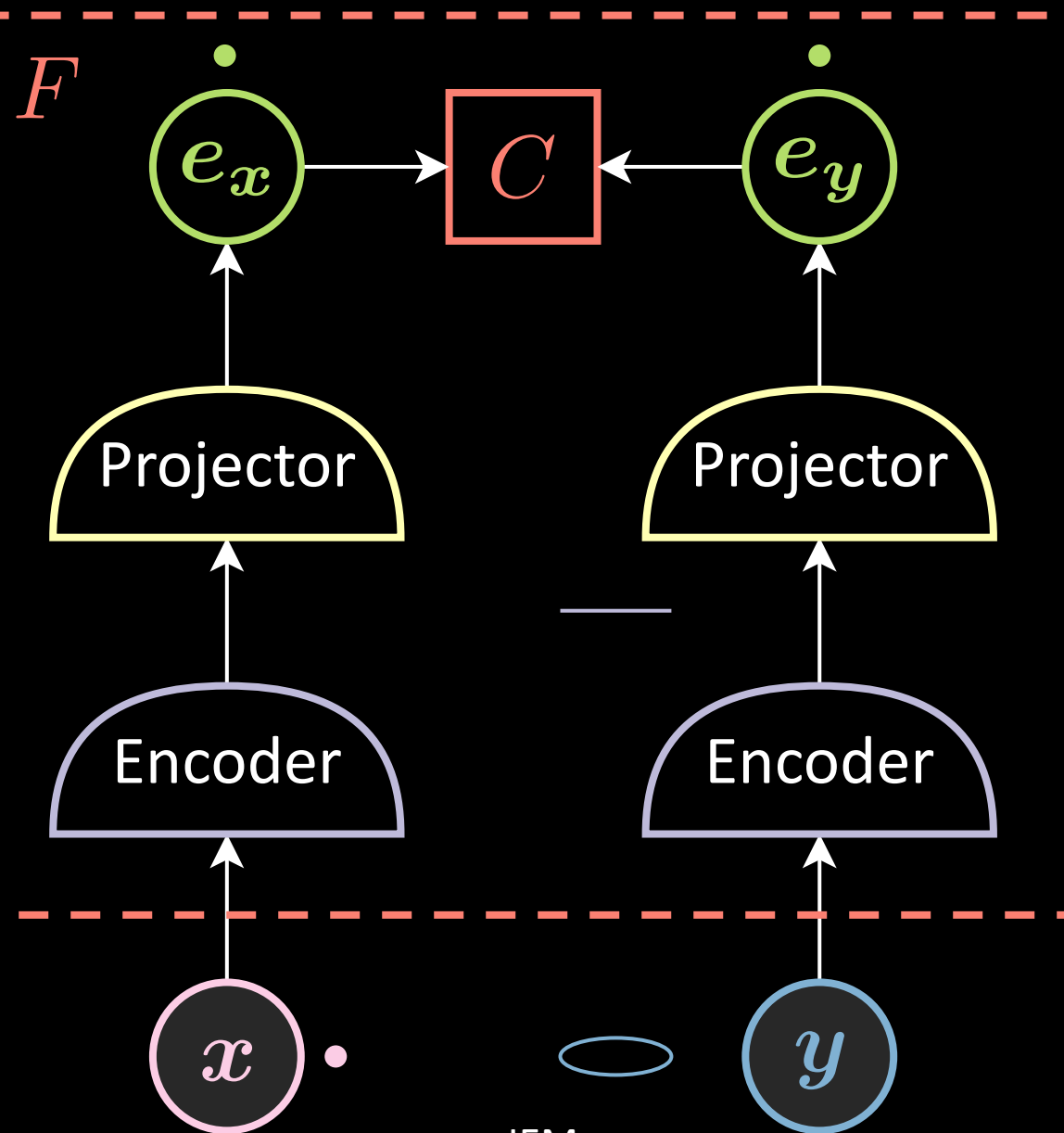
EBM architectures

1. Latent variable generative energy-based models (LVGEBMs)
2. Joint embedding methods (JEMs)

$$E(x, y, z) = C(y, \tilde{y})$$

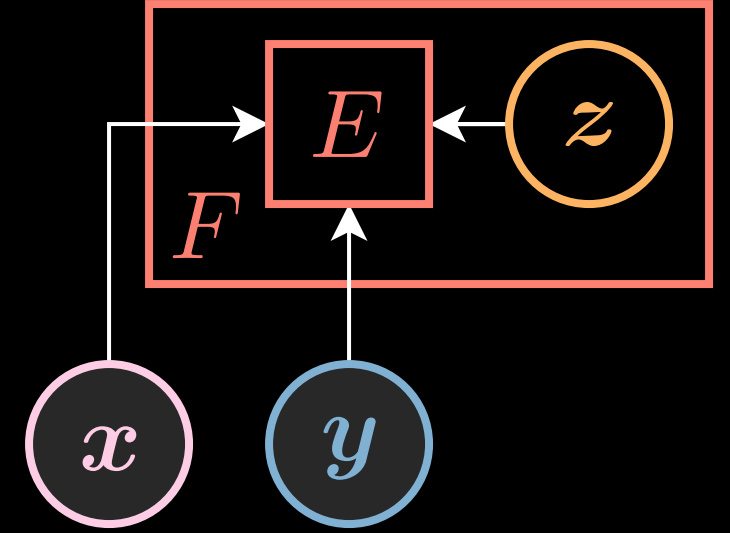


$$F(x, y) = C(e_x, e_y)$$



LVGEBM inference

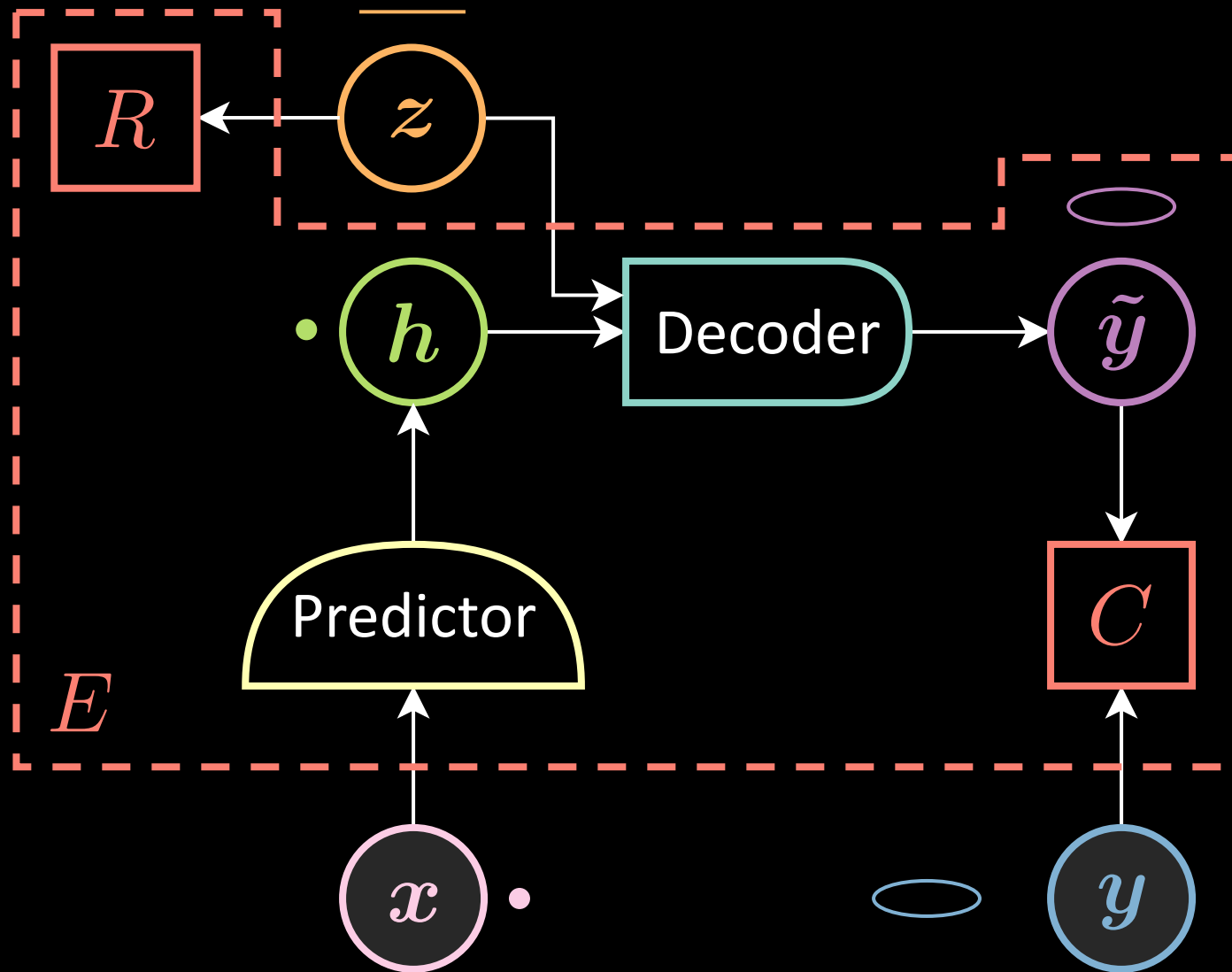
$$F_{\beta}(x, y) = \text{softmax}_{\beta} E(x, y, z)$$



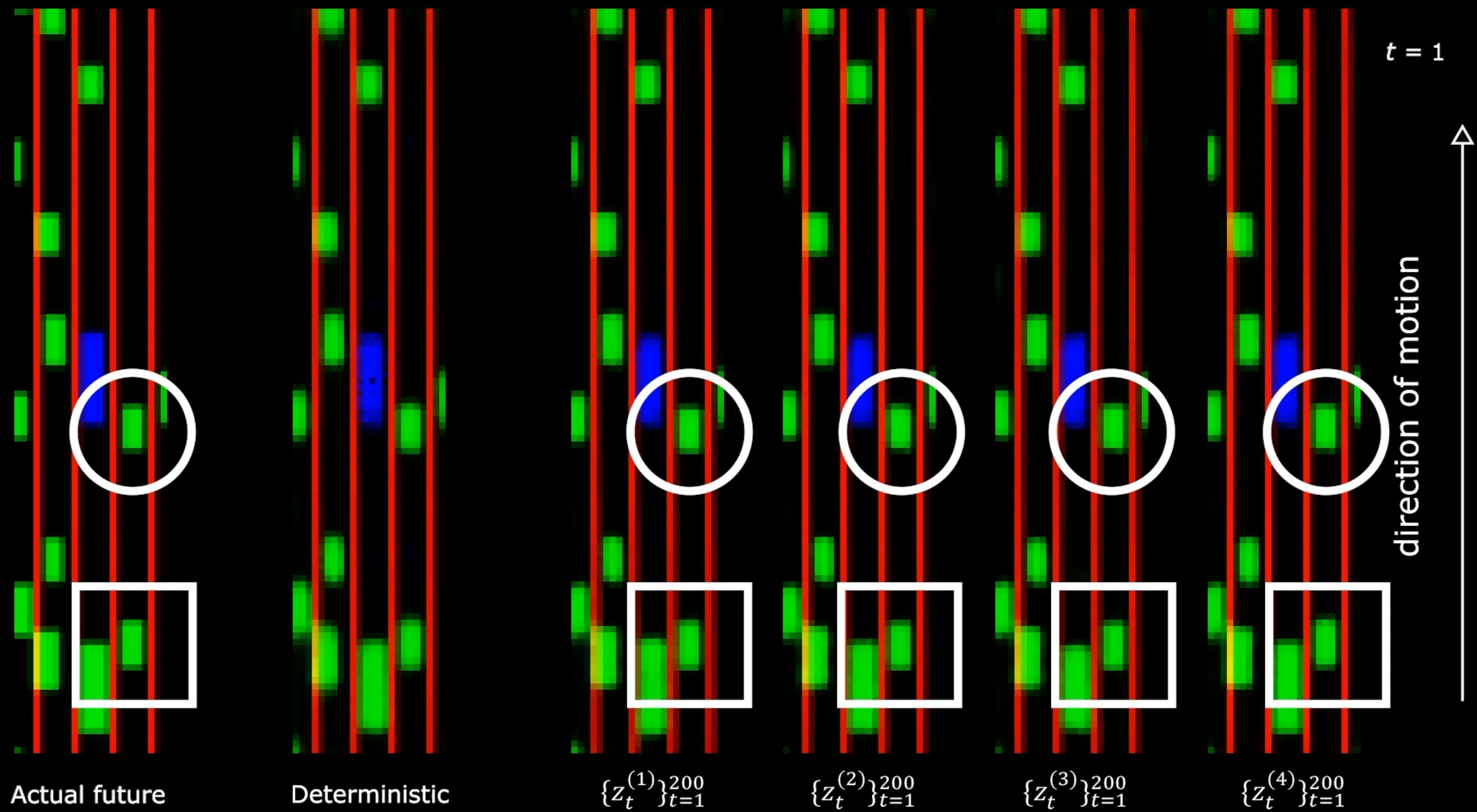
$$\text{softmax}_{\beta}(e) = -\frac{1}{\beta} \log \sum_{n=1}^N \exp(-\beta e_n)$$

$$= -\text{softmax}_{\beta}(-e) \xrightarrow{\beta \rightarrow +\infty} \min(e)$$

$$\tilde{y} = \arg \min_{y \in \mathcal{Y}} F_{\beta}(x, y)$$



$$E(x, y, z) = C(y, \tilde{y}) + R(z)$$



Training EBMs

1. Contrastive methods
2. Architectural & regularised methods

Training (I)

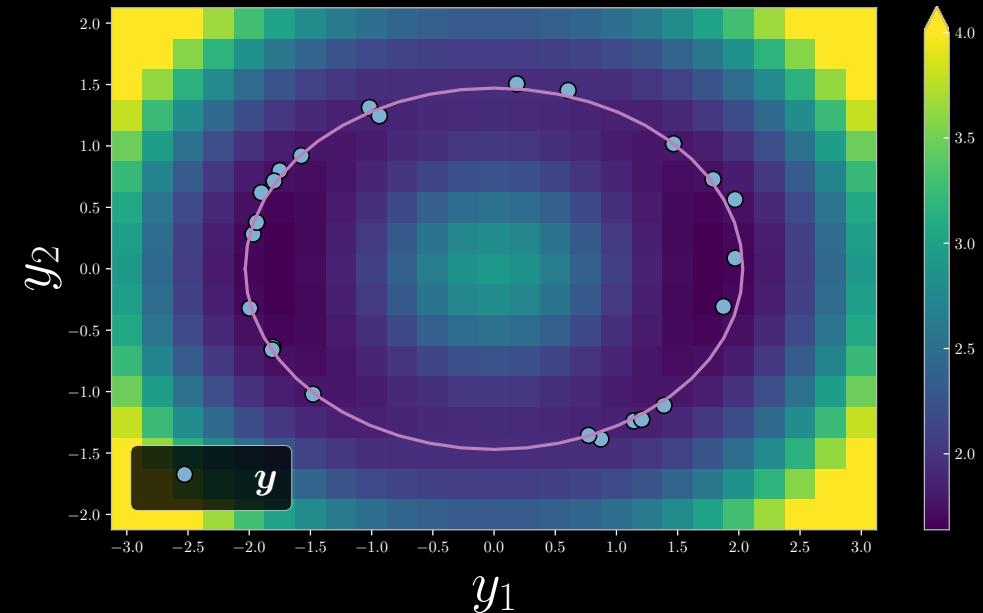
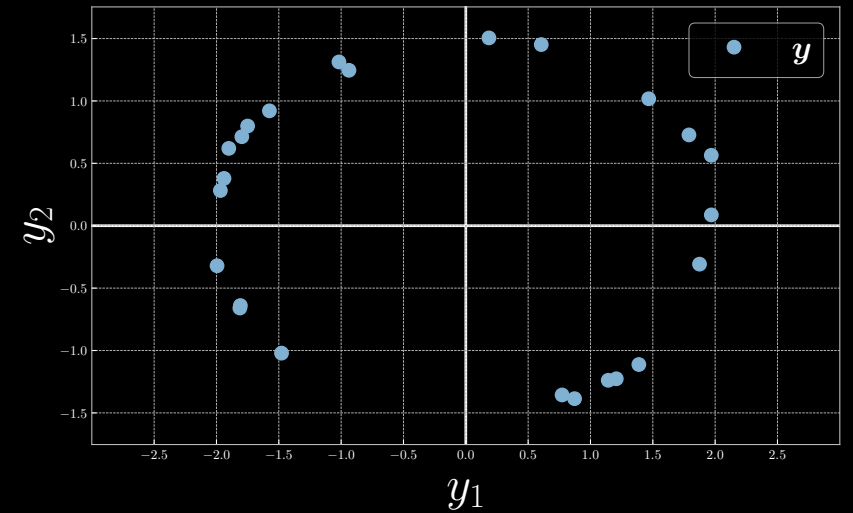
Find $F \ni F(x, y) < F(x, \mathcal{Y} \setminus y)$

1. Contrastive

Push down on $F(x, y)$
push up on $F(x, \hat{y})$

2. Architectural & regularised

Limit the volume of low energy by
construction or regularisation



Training (II)

1. Contrastive

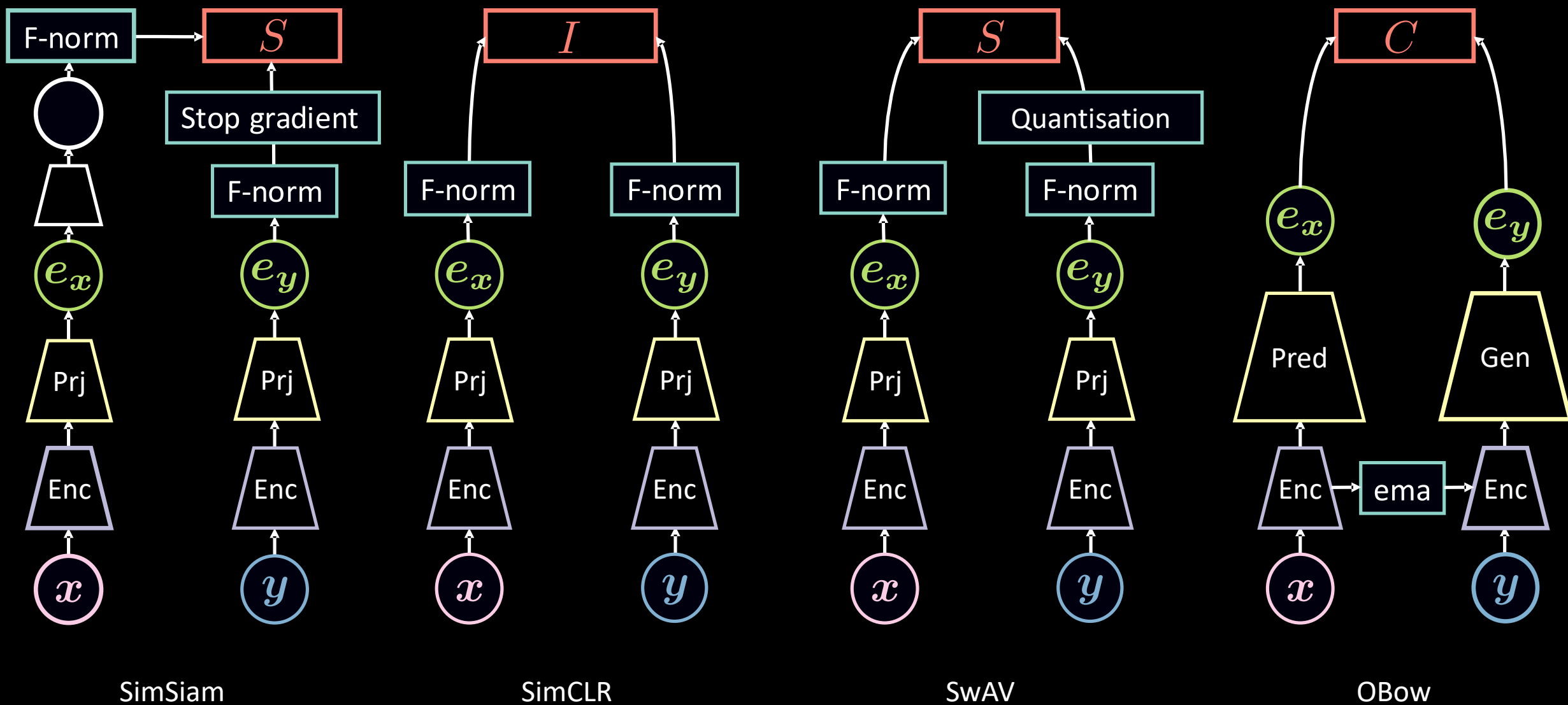
- a) Push up everywhere* (maximum likelihood)
- b) Push up at locations* (ML with MCMC, contrastive divergence, GAN, Siamese nets [PIRL, MoCo,])
- c) Off-manifold to on-manifold* (denoising & masked AE [BERT])

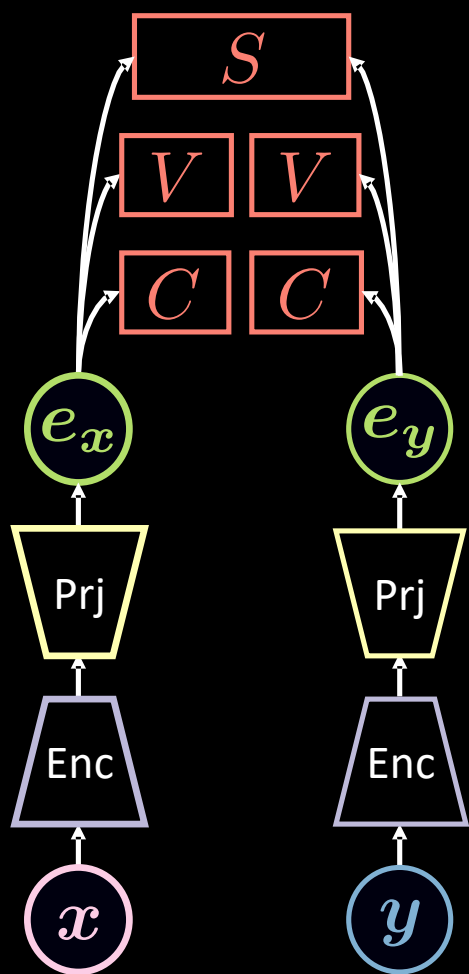
2. Architectural & regularised

- a) Upper-bound low energy volume* (PCA, K-means, GMM)
- b) Regularisation term* (sparse & contractive & variational AE, LISTA, SwAV, BYOL, VICReg)
- c) Minimise ∇ , maximise ∇^2* (score matching)

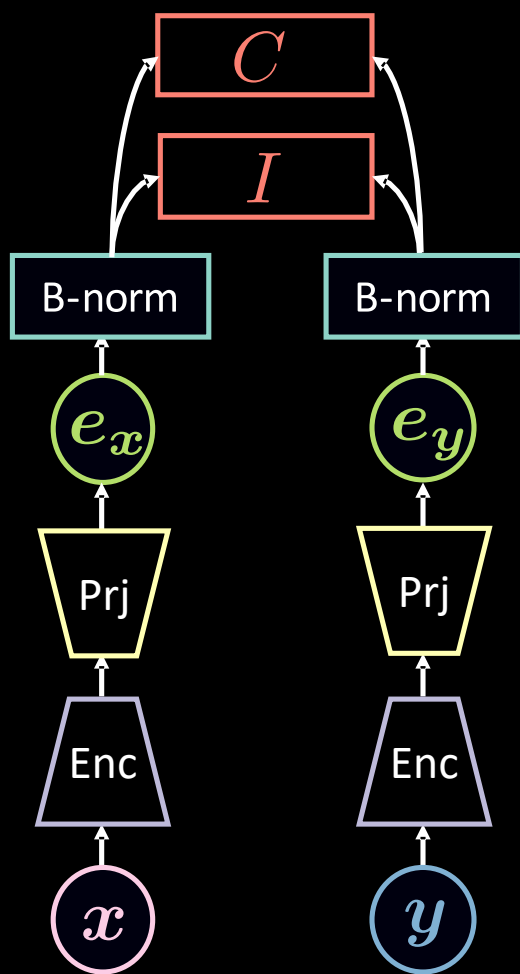
Joint Embedding Methods (JEMs)

Contrastive, clustering, distillation, and information maximisation

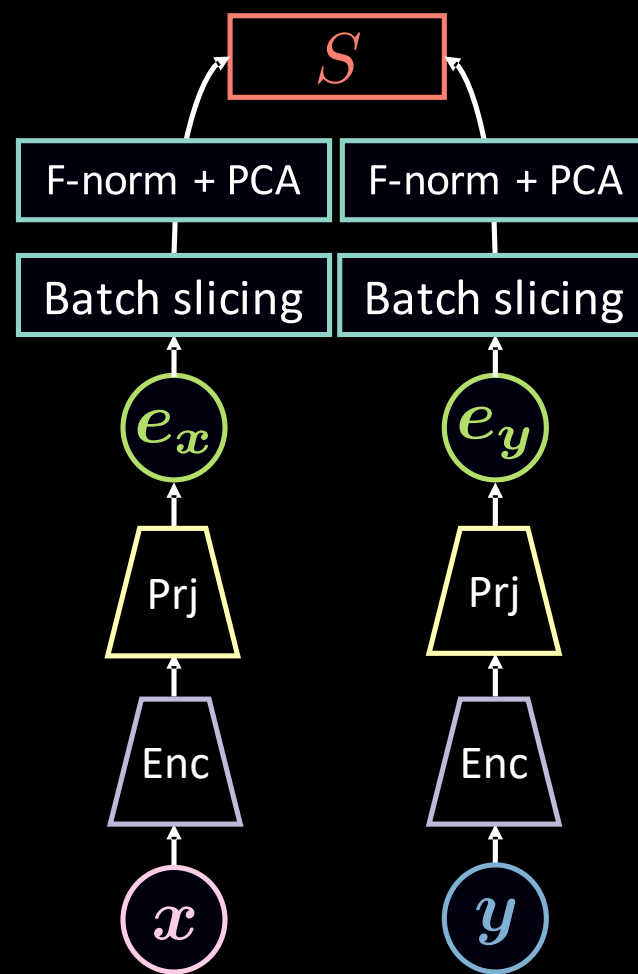




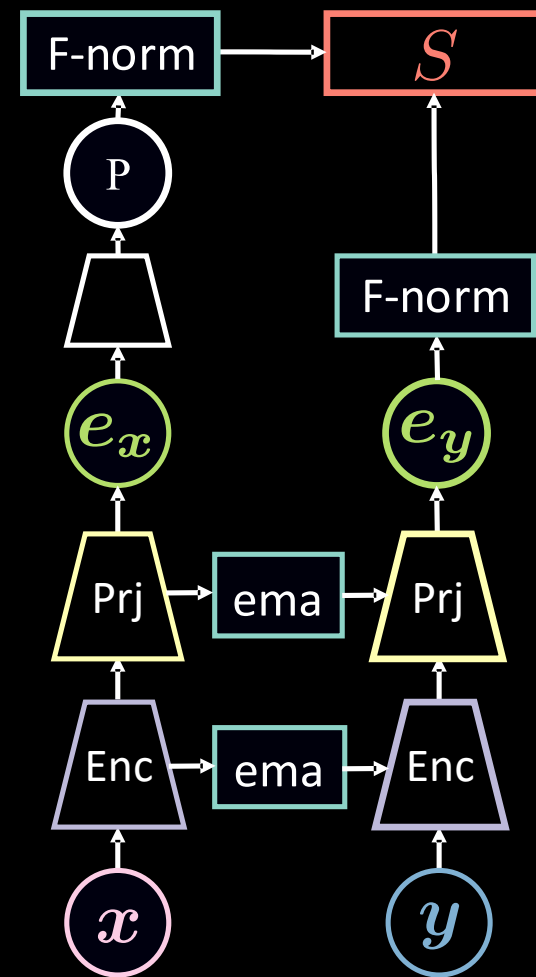
VICReg



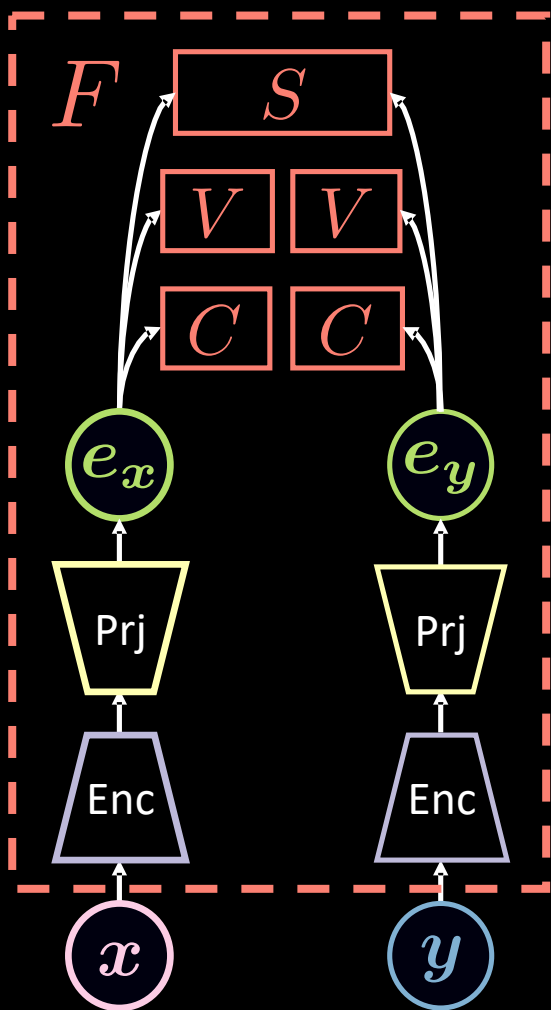
Barlow Twins



W-MSE



BYOL



VICReg

$$\mathbf{e} \in \mathbb{R}^{B \times D}, \mathbf{e}_d \in \mathbb{R}^B, \mathbf{e}^{(b)} \in \mathbb{R}^{1 \times D}$$

B: batch_size
D: emb_dimension

$$\mathring{\mathbf{e}} = \mathbf{e} - \frac{1}{B} \sum_{b=1}^B \mathbf{e}^{(b)}$$

$$V(\mathbf{e}) = \frac{1}{D} \sum_{d=1}^D \left[\gamma - \sqrt{\frac{1}{B} \|\mathring{\mathbf{e}}_d\|^2 + \varepsilon} \right]^+$$

$$C(\mathbf{e}) = \frac{\|\mathring{\mathbf{e}}^\top \mathring{\mathbf{e}}\|_F - \|\text{diag}(\mathring{\mathbf{e}}^\top \mathring{\mathbf{e}})\|^2}{D(B-1)}$$

$$S(\mathbf{e}_x, \mathbf{e}_y) = \frac{1}{B} \sum_{b=1}^B \|\mathbf{e}_x^{(b)} - \mathbf{e}_y^{(b)}\|^2$$